



First Semester 2026-27
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No. : BIO F418
Course Title : Genetic Engineering Techniques
Instructor in-Charge : Sandhya Marathe (sandhya.marathe@pilani.bits-pilani.ac.in, chamber no.: 3277-B)
Team of instructors : Mukul Joshi (mukul.joshi@pilani.bits-pilani.ac.in, chamber no.: 3277-A), Karnika (p20230410@pilani.bits-pilani.ac.in), Abha Gupta (p20240407@pilani.bits-pilani.ac.in), Rahul Bhati (p20250037@pilani.bits-pilani.ac.in), Anun Pandit (p20250440@pilani.bits-pilani.ac.in)

1. **Course description:** [as per the Bulletin]
2. **Scope and Objective:** This course aims to give the student hands-on experience of the essential techniques used in the molecular biology laboratory, with specific emphasis on DNA manipulation. Laboratory practicum will be complemented with lectures explaining the principles of the experiments being performed.
3. **Course learning outcomes:**
 - a. Learn the basics of gene cloning methods.
 - b. Gain hands-on training in the molecular biology methods.
 - c. Learn basic DNA analysis software with hands-on training.
4. **Textbooks and references:**

TB1. Nicholl, D. S. T. An Introduction to Genetic Engineering. (3rd Edition) Cambridge University Press, 2008

RB1. Brown, T.A. Gene Cloning and DNA Analysis: An Introduction. United Kingdom: Wiley-Blackwell, 2010.

RB2. Sambrook J., MacCallum P. and David Russell. Molecular Cloning: A Laboratory Manual (3rd edition, three-book set). New York, USA: CSHL Press, 2001.





5. Course Format & Weekly Workflow

There would be one theory class focusing on the principles of molecular biology techniques, with special emphasis on plasmid-based gene cloning. The theory class will be complemented by a lab practicum, where students will receive hands-on training in performing cloning and confirming the cloned product.

Week no.	Module Number	Lect./Prac. no.	Reference	Learning Outcome
1.	1. Introduction to the course and Lab orientation	1-2	-	Understanding the working principles of the lab equipment and preparing the reagents
2-3	2. DNA: isolation and analysis.	3-5	TB: 3	Understanding principles of and learning DNA/plasmid isolation and purification.
4.	3. Gene cloning: prokaryotic vectors	6	TB: 5	Learning about the different essential features prokaryotic vectors must possess for gene cloning.
4-5.	4. Gene cloning strategies I	7-9	TB: 4, 6	Learning to plan gene cloning experiments: restriction enzymes and PCR-based cloning methods.
6.	5. <i>E. coli</i> as a host for cloning	10-12	TB: 5 & notes	Knowing about the bacterial host, <i>E. coli</i> , and its versatile properties for different cloning options. Learning different transformation methods and their principles.
7.	6. Polymerase Chain Reaction	13-14	TB: 7	Understanding basic working principles of PCR and PCR for cloning.
8.	Mid-semester lab exam	15-16	NA	NA
9.	7. Gene cloning: Restriction enzymes	17-18	TB: 4	Learn about restriction enzymes and their use in cloning and mapping
10.	6. Gene cloning strategies II	19-20	TB: 4, 6	Learning to plan gene cloning experiments using different DNA-modifying enzymes, linkers, and adaptors.



11-12	8. Selection and screening procedures for cloning	21-24	TB: 8	Learning to choose an appropriate selection and screening procedure depending on the cloning strategy.
13.	9. Recombineering	25	Notes	Learning different recombineering methods.
14.	Lab comprehensive examination	26-28	NA	NA

6. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quizzes, Assignments and discussions	15%	variable	TBA	Closed/open book
Lab performance	30%	NA	Every lab	Open notes
Mid semester	10%	One week		Lab-based evaluation
Comprehensive	45%	Lab-compre (15%): 7-8 days; Theory (30%): 3 hours	1/12 FN	Closed/open book

Criterion for NC: Not attending any of the above components or 30% of the average of the top three performers

- Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
- Chamber consultation hours:** Saturday, 11 am -12 noon.
- Makeup Policy:** For a foreseen absence, a make-up request should be made to the instructor-in-charge. Reasons for unanticipated absence that qualify one for make-up include medical or similar personal emergencies only. Makeup will be given only for the Mid-semester or comprehensive exam only.

Instructor-in-charge

[BIO F418]

